Result 1 (restoring a restricted label). [Interval restriction, original label round trip]
Let I be a type. Let n, a and m be natural numbers. Assume $a + m \leq n$ (h). Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let v be a vertex in $\{0, \ldots, m + 1 - 1\}$. Let *support* be evidence identifying $A(L, a + v)$ with the elements of $L|_{[a,a+m]}$ selected by $R(L, v)$. Let x be an element of $A(L, a + v)$. Then

$$\iota(\rho(x)) = x.$$

Result 2 (restricting a restored label). [Interval restriction, surviving label round trip]
Let I be a type. Let n, a and m be natural numbers. Assume $a + m \leq n$ (h). Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let v be a vertex in $\{0, \ldots, m + 1 - 1\}$. Let *support* be evidence identifying $A(L, a + v)$ with the elements of $L|_{[a,a+m]}$ selected by $R(L, v)$. Let y be an element of $R(L, v)$. Then

$$\rho(\iota(y)) = y.$$

Result 3 (active labels correspond). [Interval restriction, value preserving equivalence]
Let I be a type. Let n, a and m be natural numbers. Assume $a + m \leq n$ (h). Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let v be a vertex in $\{0, \ldots, m + 1 - 1\}$. Then

$$(\text{there exists } e : A(L, a + v) \simeq R(L, v),\ (\text{for every } x : A(L, a + v),\ e(x) = x))\,.$$

LLM-assisted statements and proofs (gpt-6, 2026-11-10). See PROVENANCE.md. The existence result constructs support from interval bounds; the two round-trip statements take support explicitly.